

Autodesk® **Scaleform**®

Scaleform

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Autodesk® Scaleform® 4.2

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1 Scaleform 3.x

Autodesk® Scaleform® 3.0 , (AMP) ,
Scaleform 3.2 , Scaleform 3.3 ,
 , ; 1.2
Scaleform 3.3 ,

-
- MovieViews Scaleform
- ActionScript
-

1.1

Scaleform
/ Scaleform

1.1.1 (Override Allocator)

Scaleform , Gfx::System ,

1. Scaleform::SysAlloc , Alloc/Free/Realloc ,
(dmalloc),
2. Scaleform ()
“(Pause Menu) , (Memory Arena)
3. Scaleform (Scaleform::SysAllocWinAPI)
(hardware paging)

(2)

2 “ ”

1.1.2 (Memory Heap)

- Global – AMP MemReport
- MovieData – SWF/GFx
- MovieView – GFx::Movie ActionScript (sandbox),
- MeshCache –

- “ ” ,
-
- (external fragmentation)
-
- – (internal fragmentation)

“ ”

Scaleform 3.3 Scaleform::SysAlloc , ,
Scaleform 3.3 GFx::Movie ,

1.1.3 (Garbage Collection)

Scaleform 3.0 ActionScript (AS) , AS
(Memory Leak) CLIK ActionScript

Scaleform , GFx::Movie , AS
Scaleform 3.3, Movie , (Garbage Collector)
, Movie ActionScript ; , ,
Scaleform 4 “ ”

1.1.4

Scaleform “stat ID” ,
 , stat ID , stat ID ,
(AMP) , Scaleform 3.2 , AMP
AMP , [AMP User Guide](#)
Scaleform::MemoryHeap::MemReport

stat ID
Scaleform ,Visual Studio
Scaleform , Scaleform
Shipping

1.2 Scaleform 3.3 API

,Scaleform 3.3
• Gfx::Movie , 4
 ,
• SysAlloc “ malloc ”, 4K ,
512 SysAlloc
 ,
 , Scaleform::SysAlloc Scaleform::SysAllocPaged,
SysAlloc SysAllocPaged SysAllocStatic
 ,

2

1.1.1 ,Scaleform

1. SysAlloc ,
2. Scaleform ,
3. Scaleform

(2) , (1)

SysAlloc , Scaleform , Scaleform
SWF ,Scaleform Scaleform::SysAlloc::Alloc ;
Scaleform::SysAlloc::Free (Scaleform)
,
,
(compartmentalization)
2.1 SysAlloc

,
,
(level loading) ,

2.2 , Scaleform –
,Scaleform ,
(“ ”)

2.1

Scaleform ,

1. SysAlloc GetInfo Alloc Free
2. Scaleform GFX::System

Scaleform Scaleform::SysAllocMalloc ,
malloc/free ((alignment)

2.1.1 SysAlloc

Scaleform SysAllocMalloc

SysAllocMalloc

```
class MySysAlloc : public SysAlloc
{
public:
    virtual void* Alloc(UINT size, UINT align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void Free(void* ptr, UINT size, UINT align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }

    virtual void* Realloc(void* oldPtr, UINT oldSize,
                          UINT newSize, UINT align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
};
```

			MySysAlloc	SysAlloc		
GetInfo	Alloc	Free			, Scaleform	
, 16		oldSize	align	Free/Realloc	, ;	
()	Alloc/Free			Realloc		

Scaleform

Gfx::System

```
MySysAlloc myAlloc;
Gfx::System gfxSystem(&myAlloc);
```

Scaleform 3.1/3.0

Gfx::System

Scaleform

Scaleform

Scaleform

Scaleform

Gfx::System

Gfx::System

```

        , CreateMovie, , ,
        ,
        , Scaleform
        ,
        , Scaleform
        , Scaleform
        , SDK
        , Scaleform 3.1
        , SWF/GFX
        ,
        ,Scaleform::Memory::DestroyArena
        ,

```

2.2.1.2

- ```

 ,
 :

```
1. `Scaleform::Memory::CreateArena` , `SysAlloc` (
  
`;SysAllocStatic`
  

```

// pbuf1 10,000,000
SysAllocStatic sysAlloc(pbuf1, 10000000);
Memory::CreateArena(1, &sysAlloc);

```
  2. `GfX::Loader::CreateMovie / GfX::MovieDef::CreateInstance,` id
  

```

 , " " ,
 // 1
Ptr<GfX::MovieDef> pMovieDef =
 *Loader.CreateMovie(filenameStr, Loader::LoadWaitFrame1, 1);
...
// 1
Ptr<GfX::Movie> pMovie =
 *pnewMovieDef->CreateInstance(MovieDef::MemoryParams(1), false);

```
  3. `GfX::MovieDef/GfX::Movie`
  2. `GfX::ThreadedTaskManager`
  
`GfX::MovieDef::WaitForLoadFinish` ,

```
pMovieDef->WaitForLoadFinish(true);
pMovie = 0;
pMovieDef = 0;
```

```
or (if Ptr is not used):
pMovie->Release();
pMovieDef->Release();
```

3. DestroyArena , ,  
CreateArena Scaleform::Memory::ArenalsEmpty

```
// , DestroyArena
// "return *(int*)0;" .
Memory::DestroyArena(1);
```

```
SysAlloc " pbuff1 "
```

## 2.3 OS

- Scaleform OS-  
Saleform::SysAllocWinAPI – VirtualAlloc/VirtualFree API
- Scaleform::SysAllocPS3 – sys\_memory\_allocate/sys\_memory\_free  
PS3  
(OS)

WinAPI PS3 SysAlloc

```
" "
```

( ) dlmalloc 2MB

## 2.3.1 SysAllocPaged

Scaleform 3.3 SysAlloc , SysAlloc SysAllocPaged  
 Scaleform , SysAllocStatic OS  
 , SysAllocPaged, SysAlloc

SysAllocPagedMalloc

```
class MySysAlloc : public SysAllocPaged
{
public:
 virtual void GetInfo(Info* i) const
 {
 i->MinAlign = 1;
 i->MaxAlign = 1;
 i->Granularity = 128*1024;
 i->HasRealloc = false;
 }

 virtual void* Alloc(UPInt size, UPInt align)
 {
 // Ignore 'align' since reported MaxAlign is 1.
 return malloc(size);
 }

 virtual bool Free(void* ptr, UPInt size, UPInt align)
 {
 // free() doesn't need size or alignment of the memory block, but
 // you can use it in your implementation if it makes things easier.
 free(ptr);
 return true;
 }
};
```

MySysAlloc SysAlloc  
 GetInfo Alloc Free  
 ReallocInPlace

1 GetInfo() – Scaleform::SysAllocPaged::Info

2 Alloc – size align Alloc  
align GetInfo MaxAlign MaxAlign 1  
align

3 Free – Alloc size align

4 ReallocInPlace – false

[Scaleform](#)

Alloc Free GetInfo  
Scaleform SysAllocPaged  
Scaleform SysAllocPaged::Info

- MinAlign –
- MaxAlign – 0 Scaleform  
1 Alloc align
- Granularity – Scaleform Scaleform  
malloc 64K
- HasRealloc – ReallocInPlace
- SysDirectThreshold- ( )  
SysDirectThreshold, ,
- MaxHeapGranularity  
,MaxHeapGranularity alloc/free ( )  
MaxHeapGranularity 4096, 4096

MaxAlign SysAlloc

1. MaxAlign 1  
Scaleform
2. MaxAlign 0  
Alloc
3. MinAlign MaxAlign

Scaleform

GfX::System

```
MySysAlloc myAlloc;
GfX::System gfxSystem(&myAlloc);
```

## 2.4

,Scaleform “ ” , :  
(Mesh Cache) Scaleform  
API

### 2.4.1 堆 API 函数

Scaleform “ new ”

```
Ptr<GfX::FileOpener> opener = *new GfX::FileOpener();
loader.SetFileOpener(opener);
```

‘ new ’ Scaleform

GfX::System , , GfX::System  
0

GfX::MeshCacheManager  
GfX::Loader::CreateMovie() SWF/GFX  
GfX::MovieDef::CreateInstance ActionScript

Scaleform

Scaleform::MemoryHeap Scaleform  
Scaleform::MemoryHeap::CreateHeap  
Scaleform::MemoryHeap::Release  
Scaleform::MemoryHeap::Alloc Scaleform::MemoryHeap::Free

Scaleform

Scaleform::MemoryHeap

- SF\_HEAP\_ALLOC(heap, size, id)
- SF\_HEAP\_MEMALIGN(heap, size, alignment, id)
- SF\_HEAP\_NEW(heap) ClassName
- SF\_FREE(ptr)

‘ id ’

Scaleform AMP

Gfx::SpriteDef

‘ pHeap ’

```
Ptr<SpriteDef> def = *SF_HEAP_NEW(pHeap) SpriteDef(pdataDef);
```

```
Gfx::SpriteDef Scaleform Scaleform
```

```
Ptr<>
```

```
Scaleform::NewOverrideBase ‘ delete ’
```

```
delete SF_FREE
```

## 2.4.1

```
Scaleform::String Scaleform::Array<>
```

```
class Sprite : public RefCountBase<Sprite>
{
```

```
 String Name;
 Array<Ptr<Sprite> > Children;
```

```
 Sprite(String& name, ...) { ... }
```

```
};
```

```
Ptr<Gfx::Sprite> sprite = *SF_HEAP_NEW(pHeap) Gfx::Sprite(name);
sprite->DoSomething();
```



GFx::Sprite

-  
GFx::Sprite

Scaleform “ ” :

- SF\_HEAP\_AUTO\_ALLOC(ptr, size)
- SF\_HEAP\_AUTO\_NEW(ptr) ClassName

SF\_HEAP\_ALLOC SF\_HEAP\_NEW

:

```
MemoryHeap* pheap = ...;
UByte * pBuffer = (UByte*)SF_HEAP_ALLOC(pheap, 100,
GFx::StatMV_ActionScript_Mem);
Ptr<GFx::Sprite> sprite = *SF_HEAP_AUTO_NEW(pBuffer + 5) GFx::Sprite();
...
sprite->DoSomething();
...
// Release sprite and free buffer memory.
sprite = 0;
SF_FREE(pBuffer);
```

GFx::Sprite

‘ pBuffer ’ SF\_HEAP\_AUTO\_NEW

Scaleform

GFx::Sprite :

```
class Sprite : public RefCountBase<Sprite>
{
 StringLH Name;
 ArrayLH<Ptr<Sprite> > Children;

 Sprite(String& name, ...) { ... }
};
```

```
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();
```

```

Scaleform::String Scaleform::Array<> “ ”
Scaleform::StringLH Scaleform::ArrayLH<>
 “ ”
 Scaleform
 °

```

## 2.4.1 堆跟蹤程式

```

GfxConfig.h SF_MEMORY_TRACE_ALL
Scaleform::Memory SetTracer , CreateHeap DestroyHeap Alloc
Free

```

SysAllocWinAPI

```

#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"

class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
public:
 virtual void OnCreateHeap(const MemoryHeap* heap)
 {
 //...
 }

 virtual void OnDestroyHeap(const MemoryHeap* heap)
 {
 //...
 }

 virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
 unsigned sid, const void* ptr)
 {
 //...
 }

 virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
 UPInt newSize, const void* newPtr)
 {
 //...
 }

 virtual void OnFree(const MemoryHeap* heap, const void* ptr)
 {
 //...
 }
};

```

```

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static Gfx::System* gfxSystem;

class MySystem : public Scaleform::System
{
public:

 MySystem() : Scaleform::System(&sysAlloc)
 {
 Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
 }
 ~MySystem() { }
};

SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)

```

3

Scaleform 3.0 , Scaleform Player AMP HUD ,  
AMP , PC Scaleform  
AMP , [AMP User Guide](#)  
Scaleform Player , HUD UI,  
Ctrl + F5 Scaleform::MemoryHeap::MemReport

```
void MemReport (class StringBuffer& buffer, MemReportType detailed,
 bool xmlFormat = false);
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport XML  
MemReportType

- MemoryHeap::MemReportBrief – Scaleform ,

|                        |           |
|------------------------|-----------|
| Memory 4,680K / 4,737K |           |
| Image                  | 1,052     |
| Sound                  | 136       |
| Movie View             | 1,739,784 |
| Movie Data             | 300,156   |

- MemoryHeap::MemReportFull – Ctrl-F6 Scaleform Player

```
SF_MEMORY_ENABLE_DEBUG_INFO
(stat ID)
```

|                          |           |
|--------------------------|-----------|
| Memory 4,651K / 4,707K   |           |
| System Summary           |           |
| System Memory FootPrint  | 4,832,440 |
| System Memory Used Space | 4,775,684 |
| Debug Heaps Footprint    | 12,884    |
| Debug Heaps Used Space   | 5,392     |
| Summary                  |           |
| Image                    | 1,052     |
| Sound                    | 136       |
| Movie View               | 1,715,128 |
| Movie Data               | 300,156   |
| [Heap] Global            | 4,848,864 |

|                   |           |
|-------------------|-----------|
| Heap Summary      |           |
| Total Footprint   | 4,848,864 |
| Local Footprint   | 2,427,860 |
| Child Footprint   | 2,421,004 |
| Child Heaps       | 4         |
| Local Used Space  | 2,417,196 |
| DebugInfo         | 120,832   |
| Memory            |           |
| MovieDef          |           |
| Sounds            | 8         |
| ActionOps         | 80,476    |
| MD_Other          | 200       |
| MovieDef          | 36        |
| MovieView         |           |
| MV_Other          | 32        |
| General           | 91,486    |
| Image             | 92        |
| Sound             | 136       |
| String            | 31,425    |
| Debug Memory      |           |
| StatBag           | 16,384    |
| Renderer          |           |
| Buffers           | 2,097,152 |
| RenderBatch       | 5,696     |
| Primitive         | 1,512     |
| Fill              | 900       |
| Mesh              | 4,884     |
| MeshBatch         | 2,200     |
| Context           | 15,184    |
| NodeData          | 15,160    |
| TreeCache         | 13,220    |
| TextureManager    | 2,336     |
| MatrixPool        | 4,096     |
| MatrixPoolHandles | 2,032     |
| Text              | 1,232     |
| Renderer          | 13,488    |
| Allocations Count | 1,822     |

- MemoryHeap:: MemReportFileSummary -

| stat ID                        | SWF | MovieData | MovieDef | MovieView |
|--------------------------------|-----|-----------|----------|-----------|
| Movie File 3DGenerator_AS3.swf |     |           |          |           |
| Memory                         |     |           |          | 1,956,832 |
| MovieDef                       |     |           |          | 152,664   |
| CharDefs                       |     |           |          | 39,848    |

|              |           |
|--------------|-----------|
| ShapeData    | 1,716     |
| Tags         | 73,656    |
| MD_Other     | 37,444    |
| MovieView    | 196,249   |
| MovieClip    | 120       |
| ActionScript | 181,917   |
| MV_Other     | 14,212    |
| General      | 1,596,712 |
| Image        | 960       |
| String       | 2,783     |
| Renderer     | 7,464     |
| Text         | 7,464     |

- MemoryHeap::MemReportHeapDetailed -

AMP

Scaleform

|                                 |           |
|---------------------------------|-----------|
| Total Footprint                 | 4,858,148 |
| Used Space                      | 4,793,972 |
| Global Heap                     | 2,405,492 |
| MovieDef                        | 80,720    |
| Sounds                          | 8         |
| ActionOps                       | 80,476    |
| MD_Other                        | 200       |
| MovieView                       | 32        |
| MV_Other                        | 32        |
| General                         | 92,566    |
| Image                           | 92        |
| Sound                           | 136       |
| String                          | 31,276    |
| Debug Memory                    | 8,192     |
| StatBag                         | 8,192     |
| Renderer                        | 2,182,960 |
| Buffers                         | 2,097,152 |
| RenderBatch                     | 5,696     |
| Primitive                       | 1,512     |
| Fill                            | 900       |
| Mesh                            | 4,884     |
| MeshBatch                       | 2,200     |
| Context                         | 15,184    |
| NodeData                        | 15,016    |
| TreeCache                       | 13,136    |
| TextureManager                  | 2,336     |
| MatrixPool                      | 8,192     |
| MatrixPoolHandles               | 2,032     |
| Text                            | 1,232     |
| Movie Data Heaps                | 300,156   |
| MovieData "3DGenerator_AS3.swf" | 300,156   |

|                                 |           |
|---------------------------------|-----------|
| MovieDef                        | 152,664   |
| CharDefs                        | 39,848    |
| ShapeData                       | 1,716     |
| Tags                            | 73,656    |
| MD_Other                        | 37,444    |
| General                         | 141,332   |
| Image                           | 960       |
| String                          | 2,312     |
| Movie View Heaps                | 1,727,936 |
| MovieView "3DGenerator_AS3.swf" | 1,727,936 |
| MovieView                       | 195,977   |
| MovieClip                       | 120       |
| ActionScript                    | 181,645   |
| MV_Other                        | 14,212    |
| General                         | 1,467,668 |
| String                          | 471       |
| Renderer                        | 7,464     |
| Text                            | 7,464     |
| Other Heaps                     | 360,580   |
| _FMOD_Heap                      | 360,580   |
| General                         | 359,944   |
| Debug Data                      | 12,884    |
| Unused Space                    | 51,292    |
| Global                          | 51,292    |
| _FMOD_Heap                      | 4,384     |
| MovieData "3DGenerator_AS3.swf" | 5,356     |
| MovieView "3DGenerator_AS3.swf" | 33,032    |

, SF\_MEMORY\_ENABLE\_DEBUG\_INFO Scaleform ,stat ID  
stat ID

- MovieDef –
- MovieView – GfxMovieView ActionScript ,
- CharDefs – Flash
- ShapeData –
- Tags –
- ActionOps – ActionScript
- MeshCache – (Vector Tesselator) EdgeAA ,
- FontCache –

## 4

Scaleform 2.2

ActionScript

ActionScript

“ circular references ”

“ cycle references ”

◦

Code:

```
var o1 = new Object;
var o2 = new Object;
o1.a = o2;
o2.a = o1;
```

ActionScript

Flash UI

Scaleform 3.0

-

“

” Scaleform

ActionScript

“ ”

Flash

Scaleform 3.1/3.0

Scaleform

*Include/GFxConfig.h*

GFX\_AS\_ENABLE\_GC

//

**#define** GFX\_AS\_ENABLE\_GC

Scaleform



## 4.1

Scaleform 3.x ◦ Scaleform 3.3  
 MovieView

MovieDef ,

```
Movie* MovieDef::CreateInstance(const MemoryParams& memParams,
 bool initFirstFrame = true)
```

MemoryParams Gfx::MovieDef :

```
struct MemoryParams
{
 MemoryHeap::HeapDesc Desc;
 float HeapLimitMultiplier;
 unsigned MaxCollectionRoots;
 unsigned FramesBetweenCollections;
};
```

- Desc – (Granularity) (Limit) (Flags) MinAlign Scaleform
- HeapLimitMultiplier – 0...1
- MaxCollectionRoots – roots roots
- FramesBetweenCollections – roots

, Movie View

```
MemoryContext* MovieDef::CreateMemoryContext(const char* heapName,
 const MemoryParams& memParams,
 bool debugHeap)
```

```
Movie* MovieDef::CreateInstance(MemoryContext* memContext,
 bool initFirstFrame = true)
```

```

MemoryContext MovieView ,
 MovieView AMP ,
 AMP , MovieView
 ,

```

```

MemoryParams::Desc
ActionScript Desc.Limit HeapLimitMultiplier

```

128K(

Boehm-Demers-Weiser (BDW)

BDW

```

if (allocs since collect >= heap footprint * HeapLimitMultiplier)
 collect
else
 expand(heap footprint + overlimit + heap footprint *
 HeapLimitMultiplier)

```

“ collect ” ActionScript

```

HeapLimitMultiplier 0.25 25%
HeapLimitMultiplier Scaleform 25%

```

Desc.Limit

Desc.Limit

MaxCollectionRoots/FramesBetweenCollections

```

MaxCollectionRoots roots “ root ”
ActionScript ActionScript ActionScript
ActionScript ActionScript ActionScript “ obj.member
= 1 ” “ obj ” roots ActionScript
 root MaxCollectionRoots ActionScript
 MaxCollectionRoots 0

```

FramesBetweenCollections

```

“ frame ” Flash SWF 30 fps 30 2

```

|                          |         |            |              |
|--------------------------|---------|------------|--------------|
| 60                       | Advance |            |              |
|                          | 1800    | 60         | 30 fps       |
| FramesBetweenCollections |         | Desc.Limit | ActionScript |
|                          | 0       |            |              |

## 4.2 Gfx::Movie:: ForceCollectGarbage

```
virtual void ForceCollectGarbage() = 0;
```

off